

Precipitation Trends and Climate Context — Atlantic Canada

Stable baseline (~90 mm)

Avg Precip per Station

90

High extreme events

Total Extreme Events

59.5K

Peak in 2023

Wettest Year

2023

Highest in NS

Wettest Province (Avg)

NS

High variability

Monthly Precip Variability

57.7%

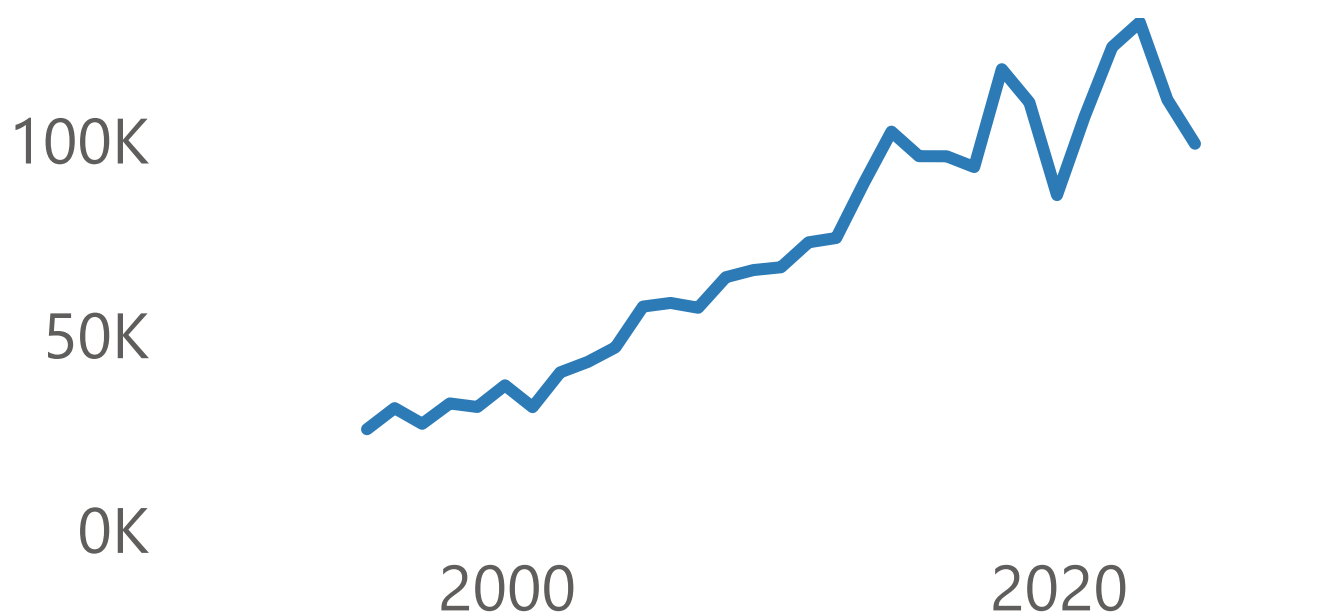
Overall volume

Total Precipitation

2.2M

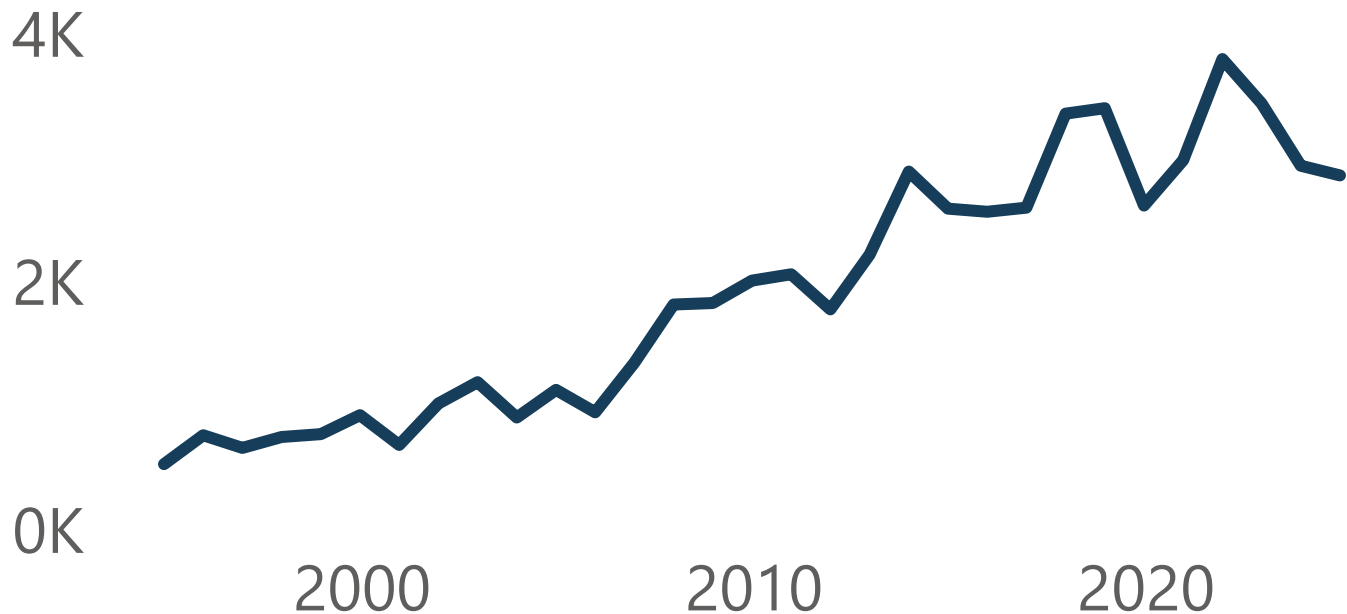
Precipitation Trend Over Time

Long-term increase with variability



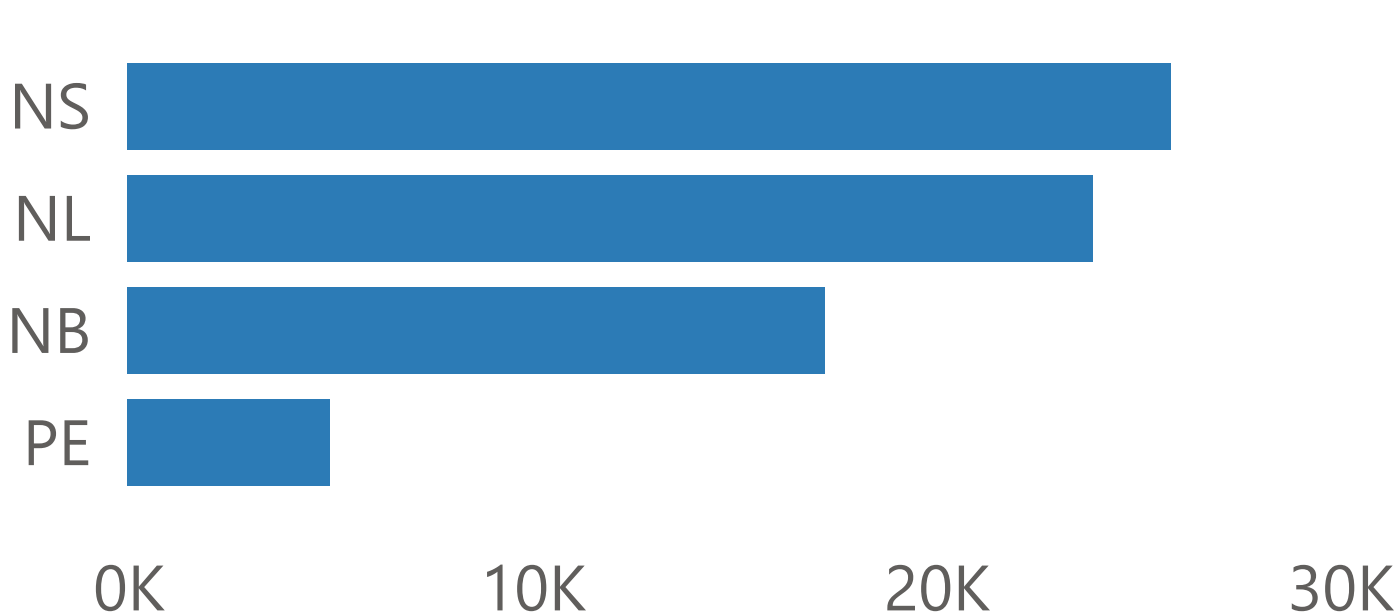
Total Extreme Events by Year

Significant increase after 2010



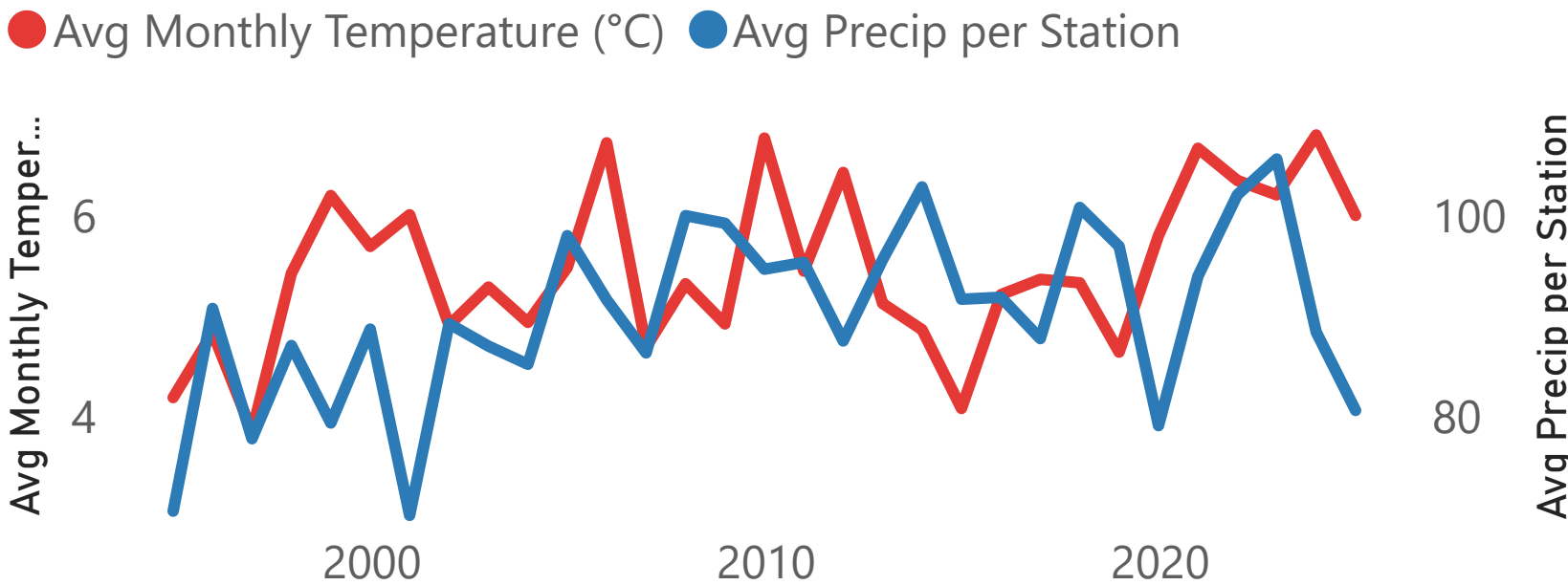
Precipitation by Province

Higher precipitation in NS and NL



Temperature vs Precipitation

No consistent relationship observed over time



Summary of Climate & Environmental Relationships

No consistent patterns observed across variables

Province	Temp	Precip	Extreme	GHG	AQI	Insight
PE	High	Low	Low	Stable	Moderate	No clear relation
NS	Moderate	High	High	Decreasing	Moderate	No clear relation
NL	Moderate	High	High	Decreasing	Moderate	No clear relation
NB	Moderate	Medium	Moderate	Decreasing	Low	No clear relation

☐ NB

☐ NL

☐ NS

☐ PE

☐ 1995

☐ 1996

☐ 1997

☐ 1998

☐ 1999

☐ 2000

☐ 2001

☐ 2002

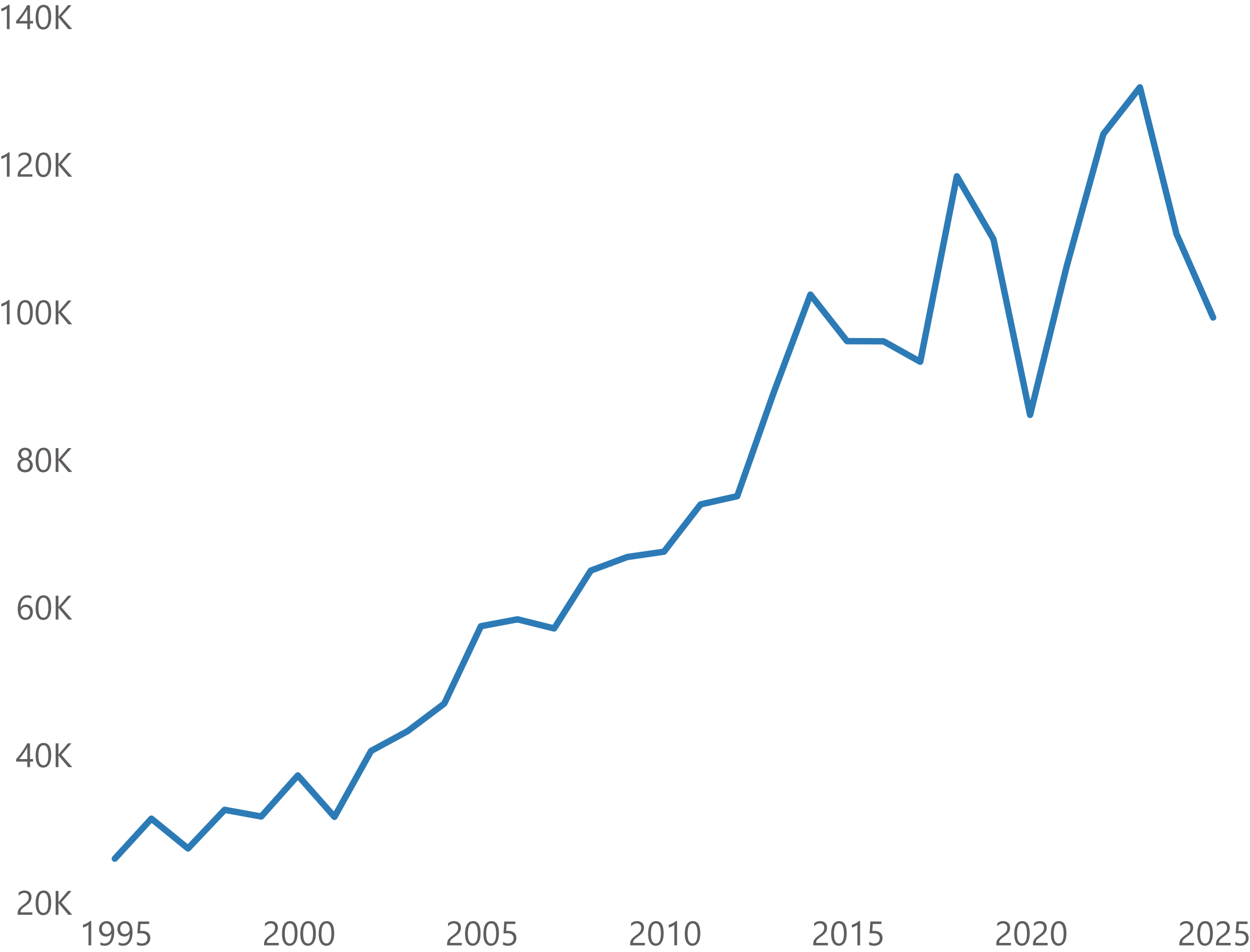
☐ 2003

☐ 2004

Precipitation Trend

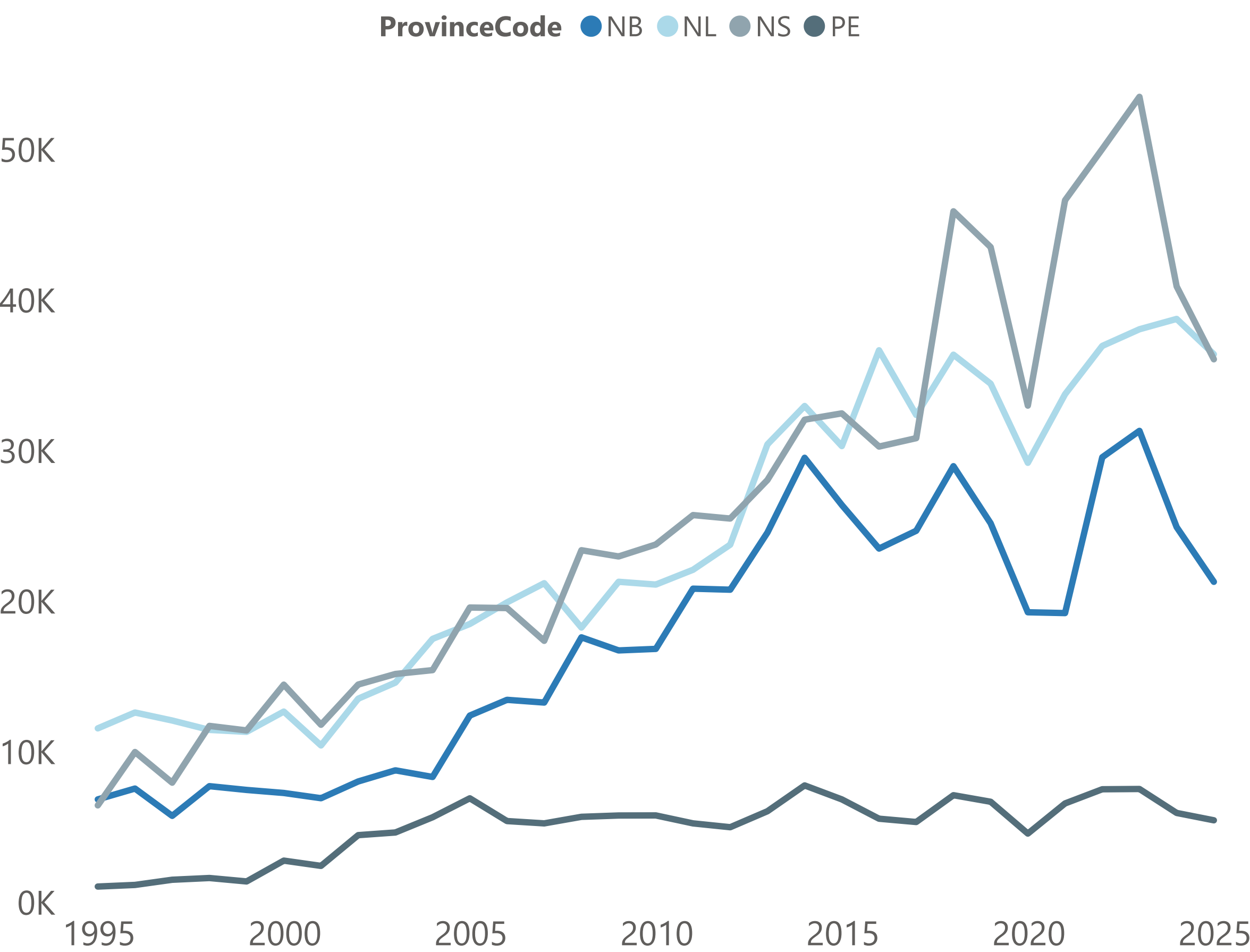
Precipitation Trend Over Time

Overall increase in precipitation over time



Precipitation Trend by Province

Overall increase in precipitation over time



Extreme Events and Regional Differences

59.5K

Total Extreme Events

35K

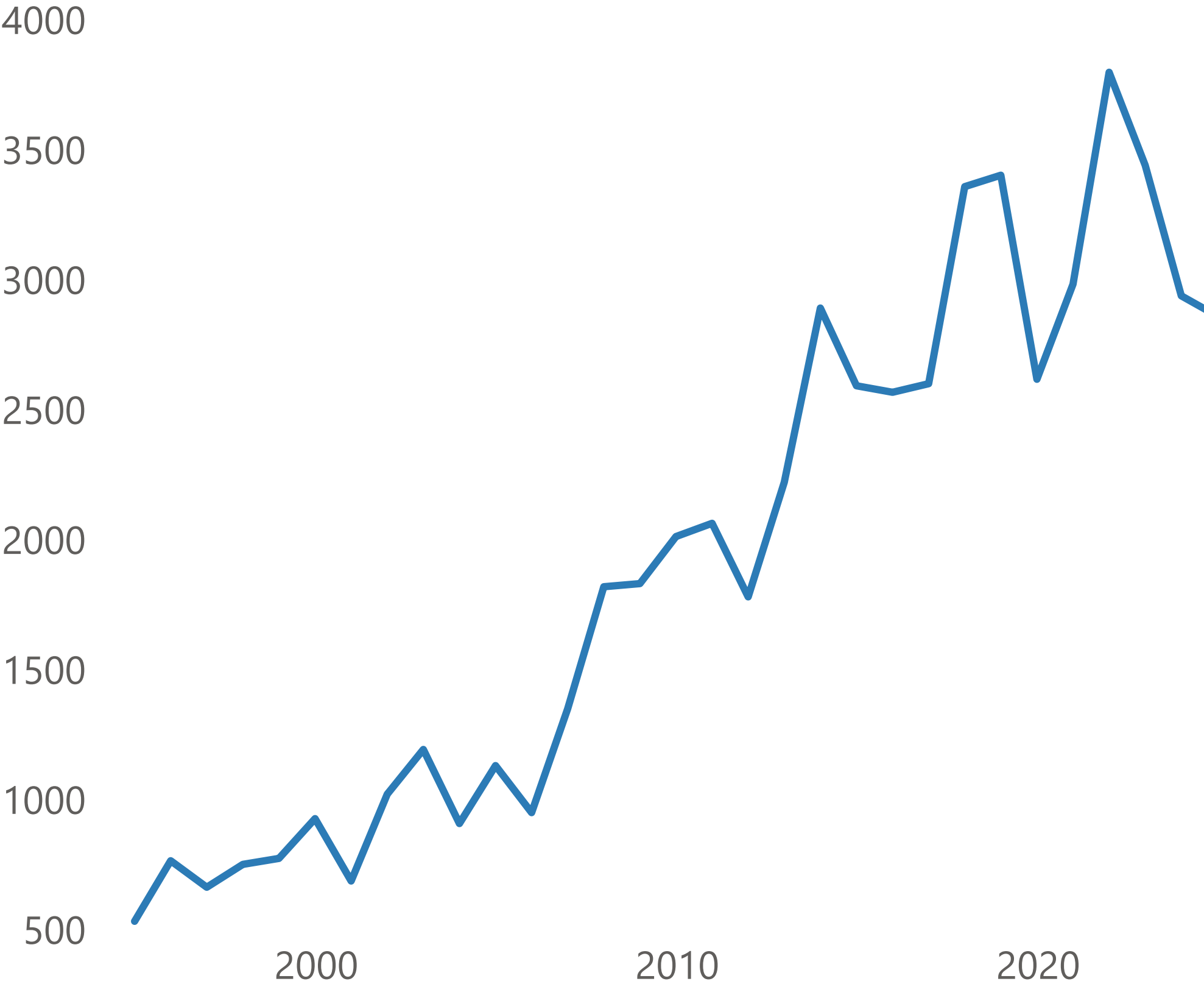
Extreme Precipitation Events

24K

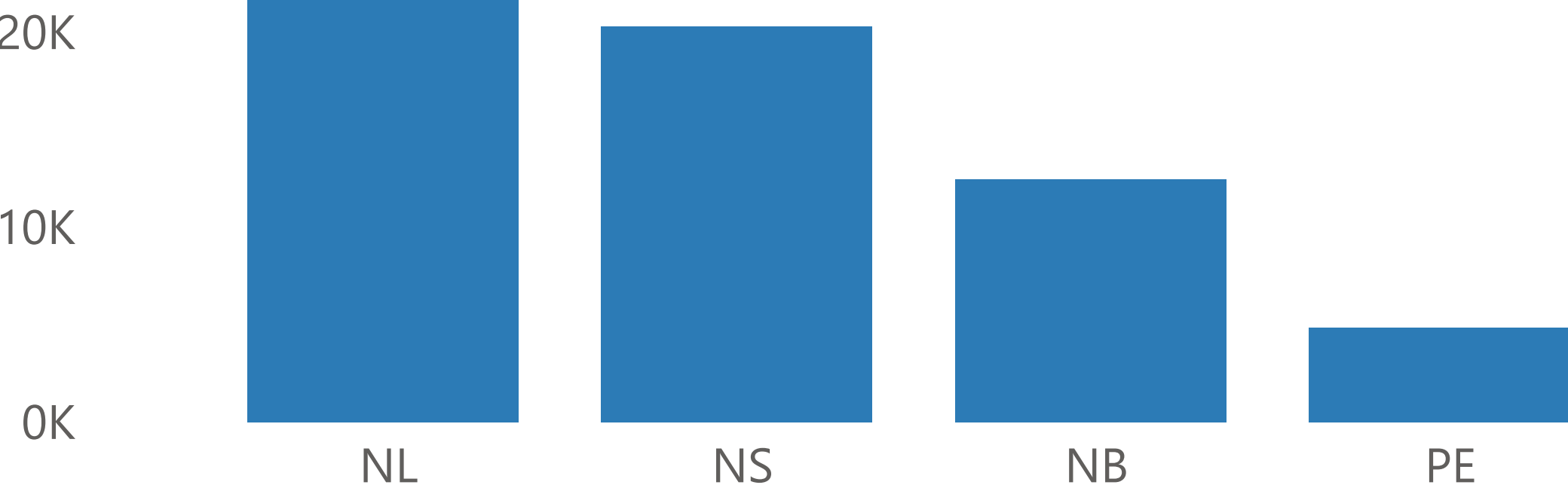
Extreme Wind Events

Total Extreme Events by Year

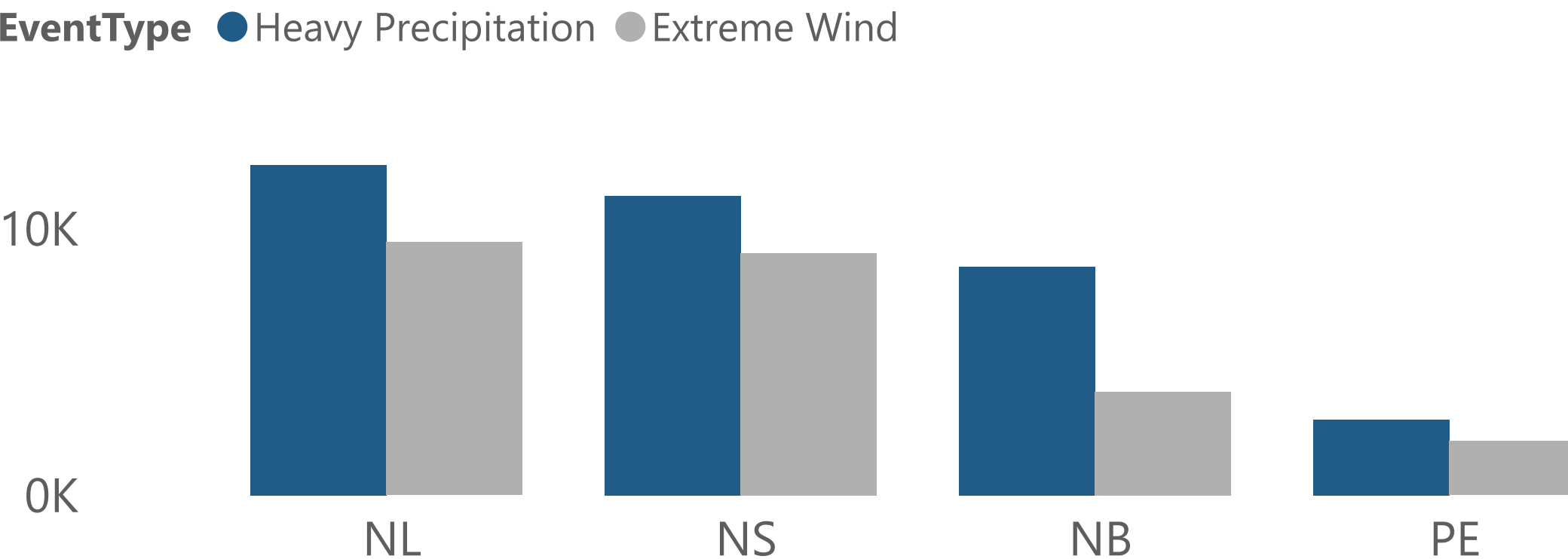
Sharp increase after 2010



Extreme Events by Province



Extreme Events by Type



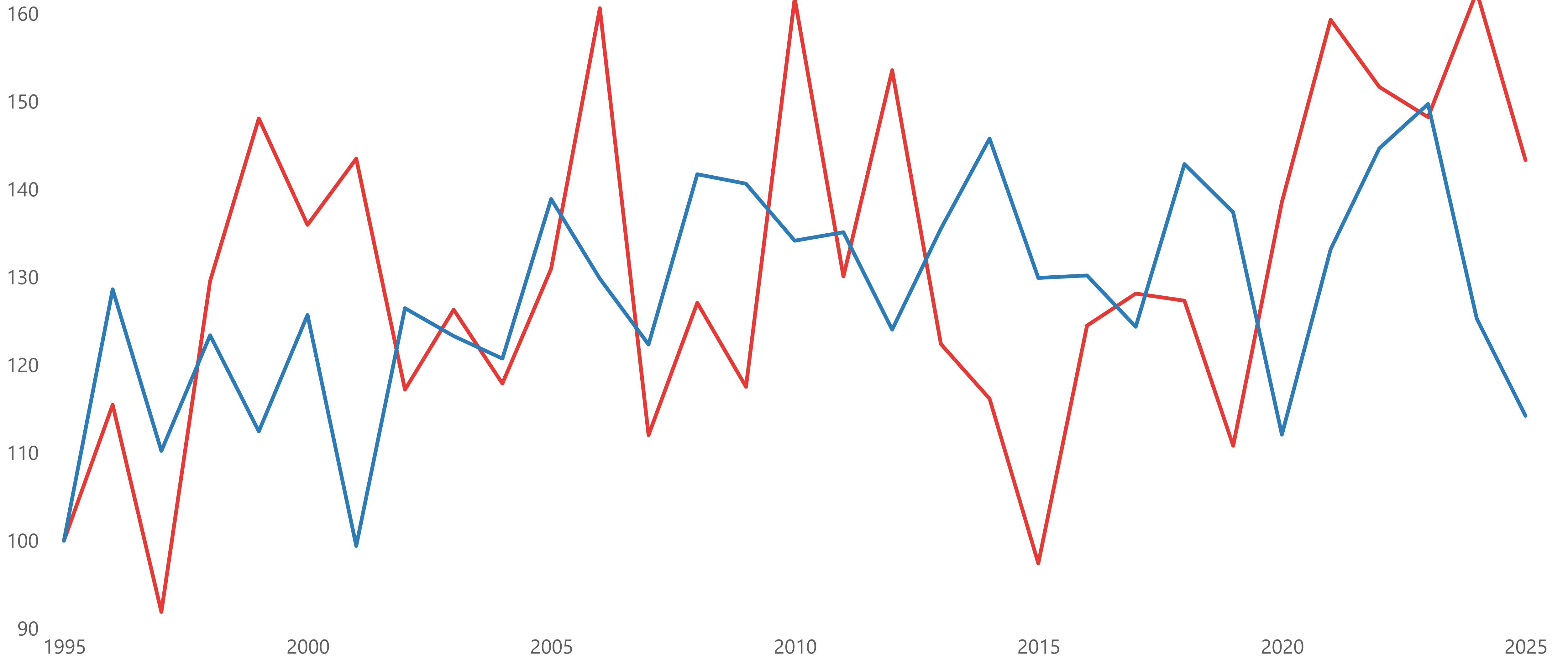
Comparing Relative Changes in Temperature and Precipitation

Similar long-term trends, but inconsistent year-to-year patterns

0.13

Correlation (long-term trend)

● Temp Index ● Precip Index



Summary of Climate & Environmental Relationships

When combining climate, emissions, and air quality indicators, no consistent relationships are observed across Atlantic provinces.

Province ▼	Temp	Precip	Extreme	GHG	AQI	Insight
PE	High	Low	Low	Stable	Moderate	No clear relation
NS	Moderate	High	High	Decreasing	Moderate	No clear relation
NL	Moderate	High	High	Decreasing	Moderate	No clear relation
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Despite a clear reduction in GHG emissions—particularly in Nova Scotia and other provinces—precipitation levels and extreme rainfall events do not show a corresponding decrease. In several cases, precipitation can remains stable or even increases as emissions decline. This suggests that precipitation patterns are not directly driven by emission trends alone.

In addition, while precipitation can contribute to temporary improvements in air quality by removing airborne pollutants, AQI levels do not display a consistent relationship with precipitation patterns across provinces. Air quality remains influenced by multiple factors, including local conditions, seasonal variation, and pollutant sources.

Overall, these findings highlight that climate and environmental relationships are complex and non-linear. Precipitation trends should therefore be interpreted with caution and analyzed within a regional context, considering multiple interacting factors rather than a single variable.